

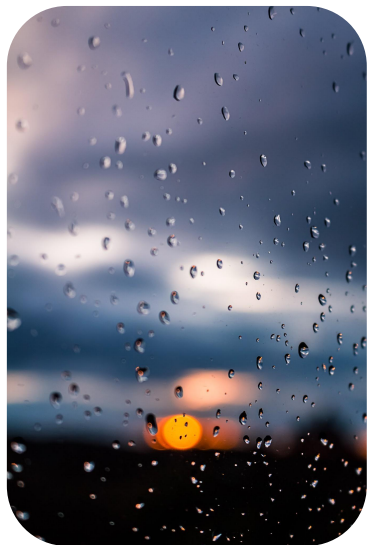


Estimating Precipitation Particle Number Concentration per Cubic Meter Using Computer Vision.

by `davy3d`

Justification

My justification for this project is that I am interested in computer vision and I like to code. I wanted to know how hard it rains outside in real time.



Vocabulary

Computer Vision: Computer vision involves how machines understand and interpret the visual world.

YOLO: YOLO stands for You Only Look Once, which is a one-stage computer vision object detection method.



Vocabulary

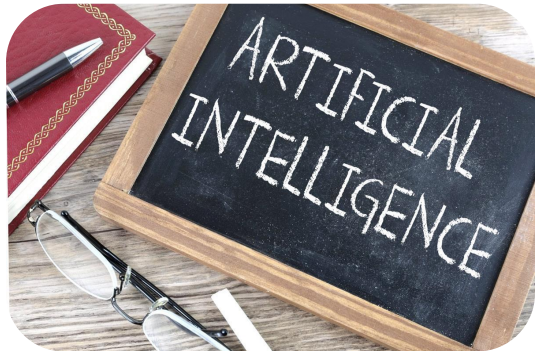
Python: Python programming language is a high-level, general-use programming language.

CSV: CSV stands for comma separated values, and is a simple way of storing information that is easy to use and manipulate.



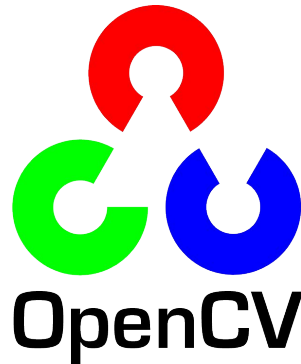
Research - Artificial Intelligence

- Artificial intelligence describes computer systems that can perform complex tasks normally done by humans.
- Machine learning is a computer learning without human input.



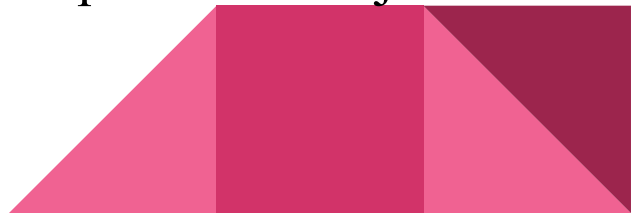
Research - OpenCV

- OpenCV is a popular computer vision library.
- An OpenCV motion detection method is to find a common background image, then subtract it from an image from the same feed. Any parts of the picture left over are detected motion.



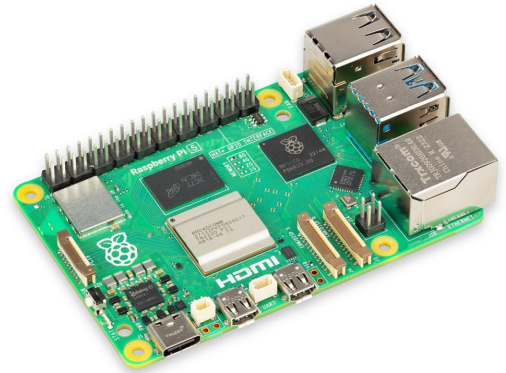
Research - Object Detection

- Different methods for object detection are Haar Cascades, Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), You Only Look Once (YOLO), and Single Shot MultiBox Detector (SSD).
- The YOLO object detection method is a convolutional neural network that is able to detect, classify, track, and estimate the pose of an object.



Research

- The Raspberry Pi is small, lightweight computer.
- In rain, raindrop concentration usually ranges from 100 to 1,000 drops per cubic meter.



Objective

The objective of this project is to use a Raspberry Pi and Python to detect rain or snow, namely precipitation, and detect the density of the raindrops, or particles, using computer vision.



Materials

- 2 Cameras
- 1 Raspberry Pi computer
- 1 Flat platform
- 1 Surrounding rig
- Python programming language
- 1 3 meter USB cable
- 1 Laptop
- 1 USB hub
- 2 Garden floodlights



My Platforms and Rigs



This is my flat platform.



This is my surrounding rig.



Procedure

1. I collected the materials.
2. I built the rigs.
3. I put the rigs on a shed roof.
4. I set up the Raspberry Pi and connected it to the cameras and the network.
5. I wrote a Python script to take pictures to train my object detection model and ran it.



Procedure

6. I wrote my Python program to use the cameras to detect motion with one camera and precipitation with the other. The motion detection detected motion and counted any areas of motion as a raindrop.
7. I trained my YOLO object detection model with pictures of precipitation that I took with my program on my flat platform.
8. I ran my program all day and night for eleven days and wrote the output to a csv file.



Pictures

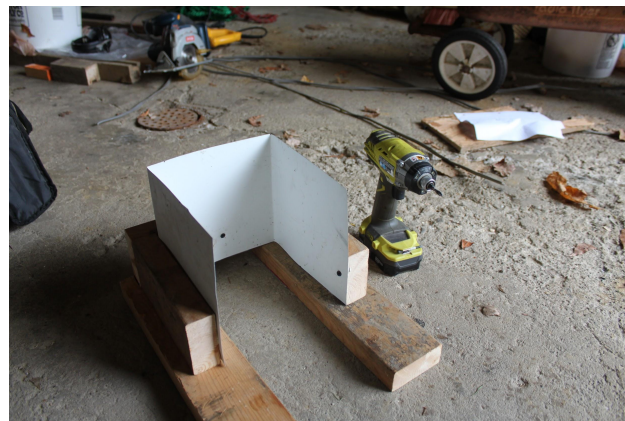
#1



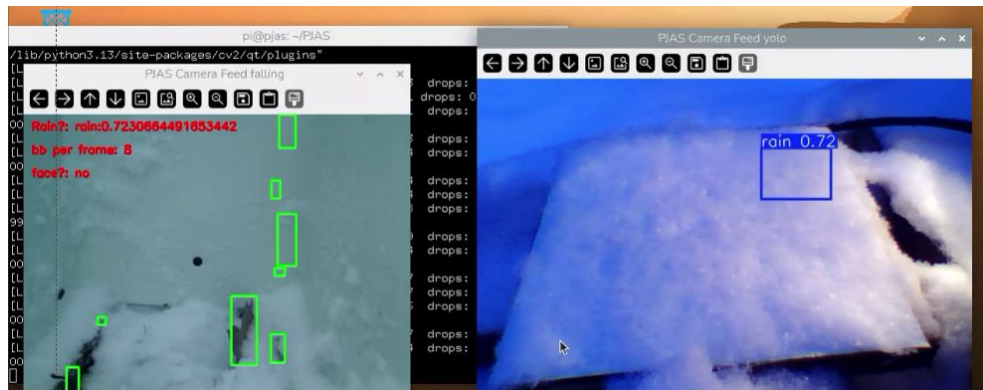
#2



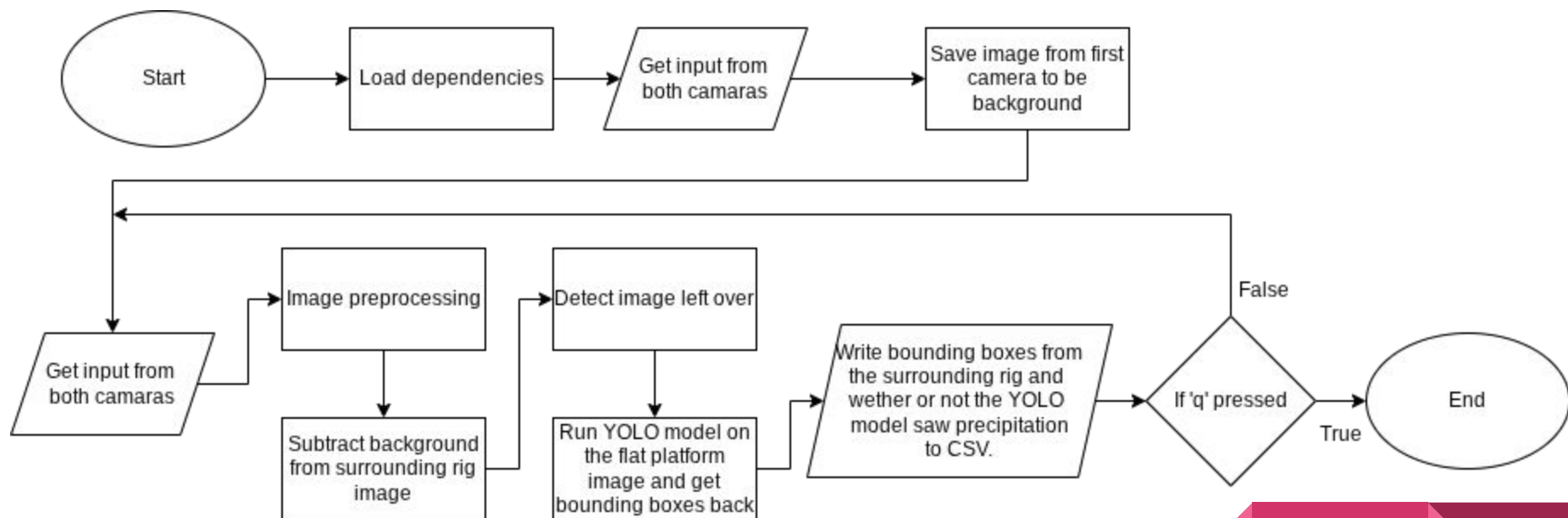
#4



#3



Flowchart

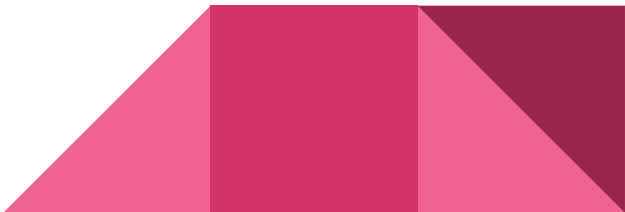


YOLO Object Detection Logic

```
1 results = rainmodel.predict(source=frameyolo, stream=True, conf=0.7, verbose=False)
2 confidence = ''
3 class_name = ''
4 class_name_conf = ''
5 for result in results:
6     # The 'plot()' method automatically draws
7     # the bounding boxes and labels on the frame
8     yolo_frame = result.plot()
9     boxes = result.bboxes
10    for box in boxes:
11        class_id = int(box.cls[0])
12        class_name = str(rainmodel.names[class_id])
13        confidence = str(float(box.conf[0]))
14        class_name_confidence = class_name_confidence + class_name + ':' +
                                + confidence + ' ',
```

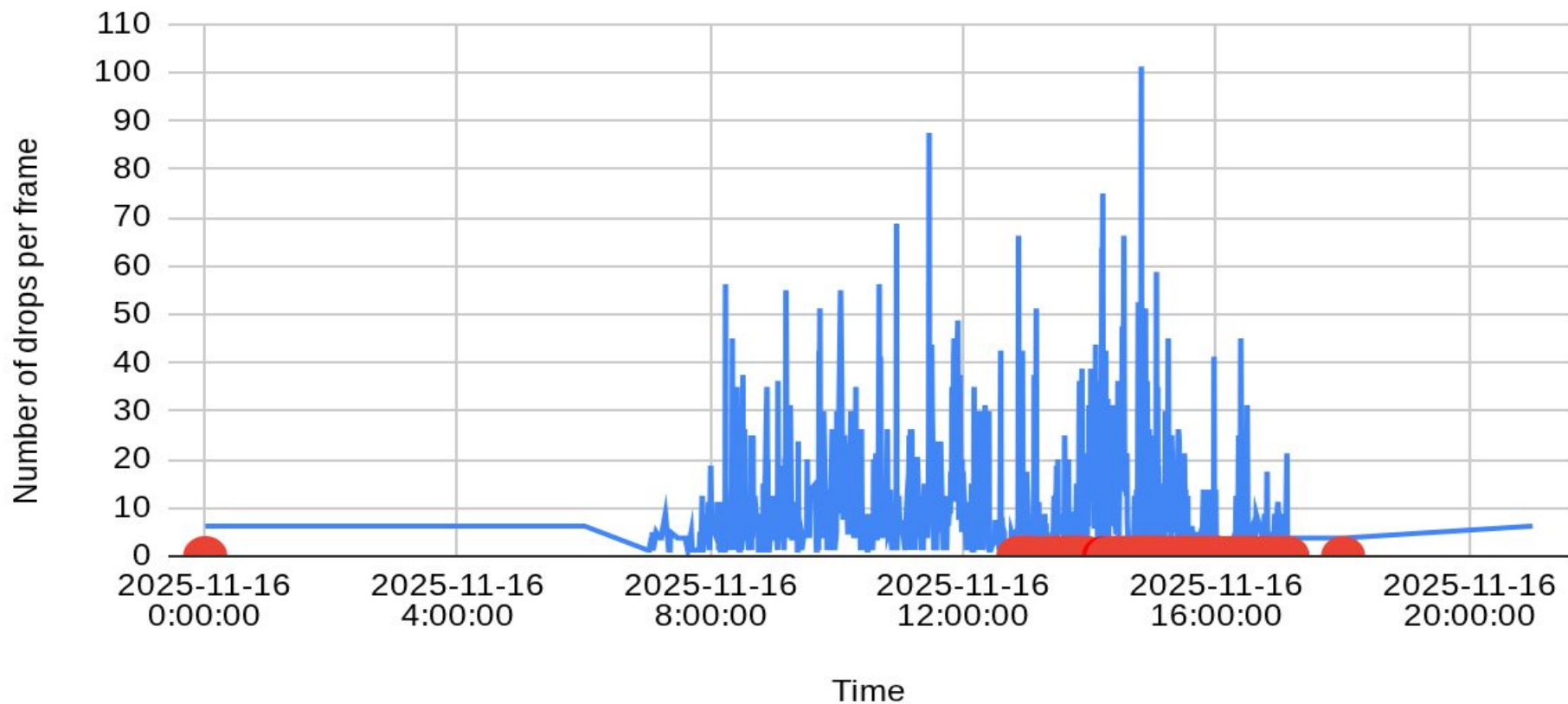
Example Output:

class_name_confidence =
'rain:0.7154446244239807'



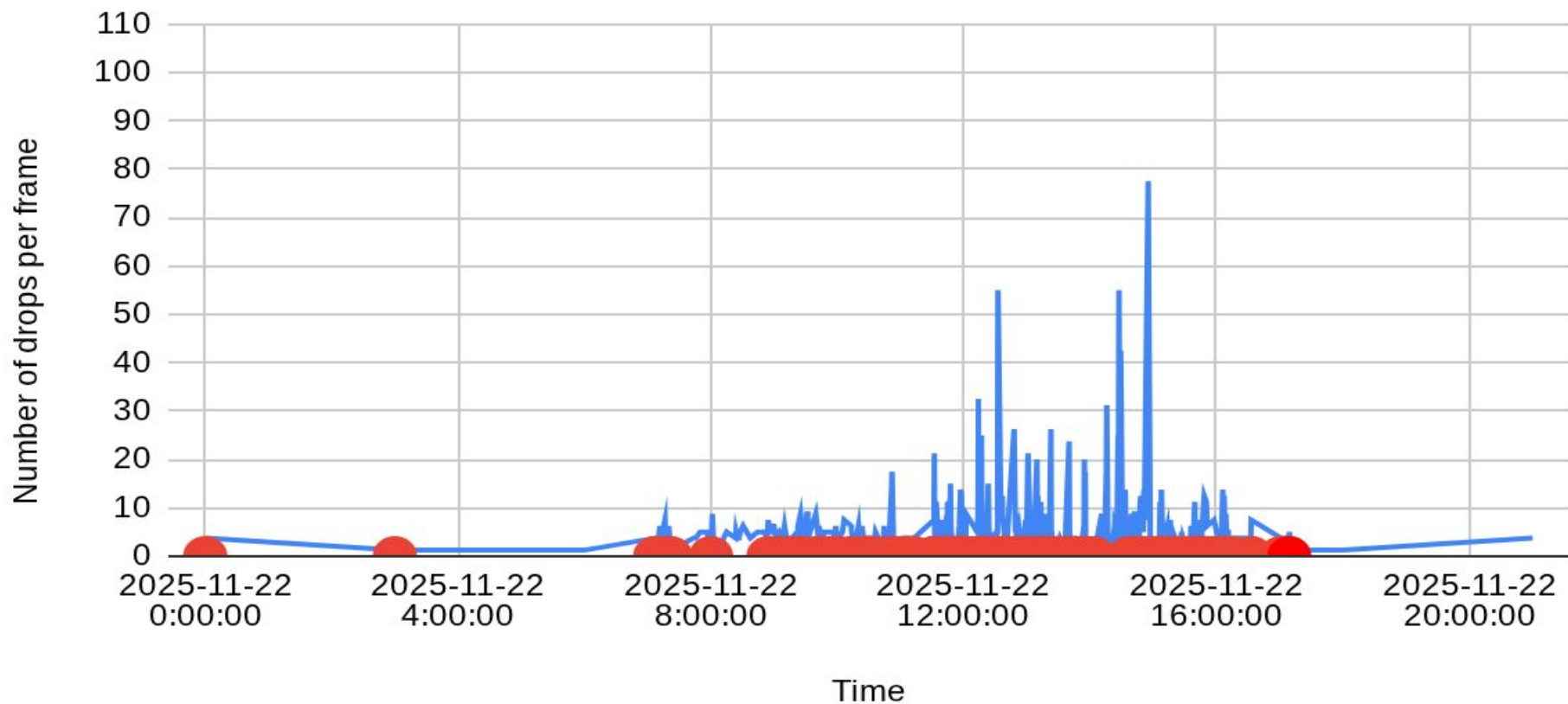
2025-11-16

Number of Drops per Frame YOLO Detected Precipitation on Detection Surface



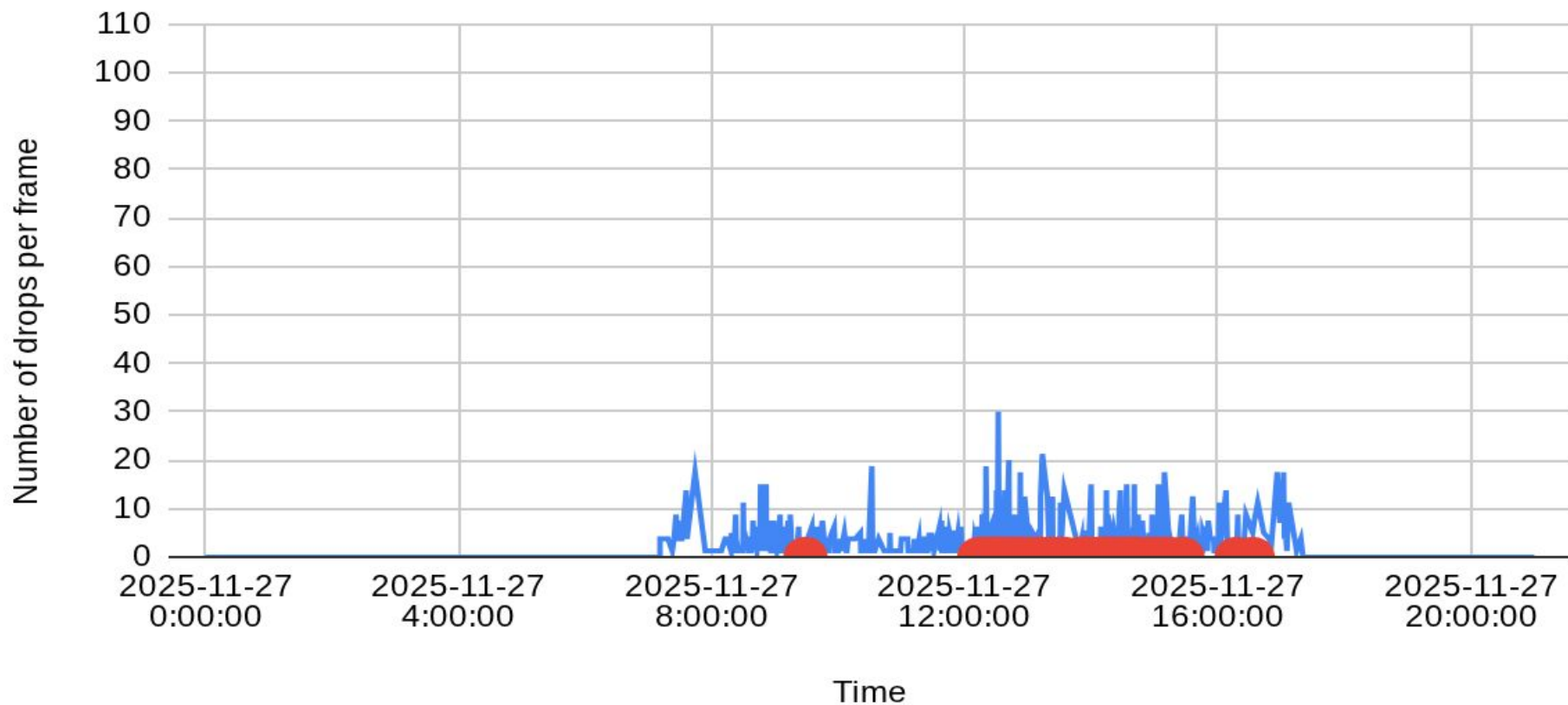
2025-11-22

Number of Drops per Frame YOLO Detected Precipitation on Detection Surface



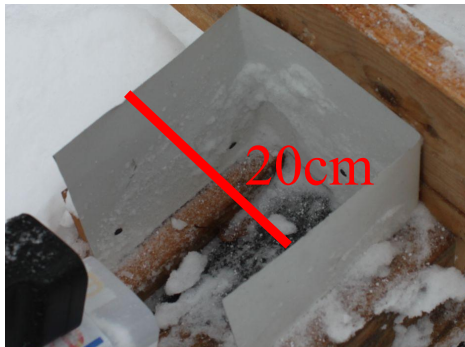
2025-11-27

Number of Drops per Frame YOLO Detected Precipitation on Detection Surface



Conclusion

In conclusion, I was able to complete my objective. I was able to get output from my program that was reasonable. It was able to detect when precipitation was on my platform and the density of the precipitation. The smallness of the raindrop detecting area may have affected the accuracy of the data, though.



Importance

- This is important because people can know the density of the falling raindrops, or how hard, it is raining outside.
- This could also be a good start on a more complicated program.



Furthering my Project

- I would enlarge the raindrop detecting area so that I could have more accurate results.
- I would train my YOLO model to detect hail, so that it could warn people when it was hailing.
- I would also make a website that people could visit that would show the current weather and the weather history.



Thank you

My Dad

Wilson Engineering
Services

My Teachers

My Siblings

My Classmates

You!!

THANK
YOU!



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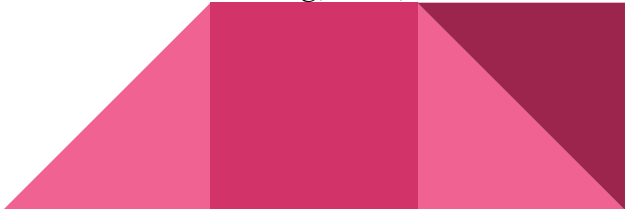
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